

MINI PROJECT REPORT

on

SATHash

by

Sahil Powar *123A9035*

Pranav Sukale *123A9054*

Swayam Sawant *123A9055*

Tanay More *123A9056*

UNDER THE GUIDANCE OF

Prof. Gilu Abraham

Submitted in Partial fulfilment of the requirements of the degree

BACHELOR OF ENGINEERING IN
ARTIFICIAL ENGINEERING AND MACHINE LEARNING



DEPARTMENT OF ARTIFICIAL INTELLIGENCE
AND MACHINE LEARNING ENGINEERING
SIES GRADUATE SCHOOL OF TECHNOLOGY (Autonomous)

NERUL, NAVI MUMBAI – 400706

AY: 2025-2026

Certificate

This is to certify that the project entitled “**SATHash**” is a bonafide work carried out by the following students of Third Year in Artificial Intelligence and Machine Learning.

Sr.no	Name	Roll No.
1	Sahil Powar	123A9035
2	Pranav Sukale	123A9054
3	Swayam Sawant	123A9055
4	Tanay More	123A9056

The report is submitted in partial fulfillment of the degree course of Bachelor of Engineering in Artificial Intelligence and Machine Learning, of University of Mumbai during the academic year 2025 – 2026.

Dr. Varsha Patil
Head of Department

Prof. Gilu Abraham
Supervisor

Dr. K Laskshmisudha
Principal

Project Report Approval

This project report entitled **SATHash** by the following student is approved for the degree of **Bachelor of Engineering in Artificial Intelligence and Machine Learning**.

Sahil Powar	123A9035
Pranav Sukale	123A9054
Swayam Sawant	123A9055
Tanay More	123A9056

Name of Internal Examiner: _____

Signature: _____

Name of External Examiner: _____

Signature: _____

Date:

Place:

Declaration

I declare that this written submission represents my ideas in our own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated, or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Name	Roll No.	Signature
Sahil Powar	123A9035	
Pranav Sukale	123A9054	
Swayam Sawant	123A9055	
Tanay More	123A9056	

Date:

Contents

Abstract	vii
Acknowledgement	ix
List of Figures	x
List of Abbreviations	xi
1 Introduction	1
1.1 Introduction	1
1.2 Motivation	2
1.3 Problem Statements and Objectives	3
1.4 Organization of the report	5
2 Literature Survey	7
2.1 Research Gaps	9
2.2 Mini Project Contribution	10
3 Proposed System	12
3.1 System Architecture	12
3.2 Architecture of Proposed System	14
3.3 Details of Hardware and Software	15
4 Design and Methodology	17
4.1 Design details	17
4.2 Methodology	18
4.3 Algorithm Implementation	20
5 Result and Discussions	23

5.1	Model Implementation	23
5.2	Results and Discussions	25
6	Conclusion	27
6.1	Conclusion	27
6.2	Future Scope	28
	Bibliography	30

Abstract

SATHash is a self-supervised deep learning-based satellite image retrieval system designed to efficiently identify and retrieve visually similar satellite image patches from large-scale multi-spectral geospatial datasets. The system operates on the BigEarthNet-S2 dataset comprising Sentinel-2 satellite images with 10 spectral bands, capturing rich land-cover information including vegetation health, water content, and soil composition that cannot be represented by standard RGB imagery. The system addresses the challenge of handling high-resolution geospatial imagery by transforming images into compact and discriminative binary representations using a custom deep neural architecture — SpectralHashNetv6 — trained entirely from scratch using self-supervised contrastive learning, specifically the NT-Xent loss from the SimCLR framework. Unlike conventional retrieval systems that depend on labeled data, SATHash requires no annotations during training, making it highly scalable to real-world unlabeled satellite archives. The architecture features a four-stage residual encoder where each stage employs Residual-BlockProj units combining Group Normalization, skip connections, and CBAM attention — a dual-branch mechanism applying both channel-wise Squeeze-and-Excitation and spatial attention to selectively emphasize informative spectral and spatial features. This encoder processes 10-band inputs of spatial dimension 120×120 , progressively reducing to a $256 \times 8 \times 8$ feature map, followed by a projection head mapping features into a normalized 128-dimensional latent space, and a hash head with GELU activations that produces compact 64-bit binary hash codes via a differentiable Tanh-Sign quantization mechanism. Group Normalization is used throughout in place of Batch Normalization, preserving per-image statistical uniqueness and preventing hash collapse. The model is trained with a composite loss comprising NT-Xent contrastive loss, quantization loss, and independence loss, with gradient clipping, AdamW optimization, and cosine annealing scheduling for stable training.

The generated binary codes are indexed using FAISS with Hamming distance, enabling millisecond-level retrieval at scale. A precomputation strategy is employed where binary hash codes are

stored in advance using NumPy and JSON, significantly reducing query time during inference. Evaluation on a proper query-database split yields 99.4 percent unique hash codes, fully balanced 64-bit activations, a mean Hamming distance of 32, a Precision@5 of 72.6 percent, along with strong mAP and NDCG@10 scores, demonstrating robust semantic retrieval performance. An interactive Streamlit-based frontend provides real-time querying, result visualization, and analytics, making the system suitable for applications such as land-use analysis, urban planning, environmental monitoring, and disaster assessment.

Acknowledgement

We would like to express our thanks to the people who have helped us the most throughout our project. We are grateful to our guide **Prof. Gilu Abraham** for nonstop support for the project.

A special thanks goes to each other who worked together as a team in completing the project, where we all exchanged our own interesting ideas, thoughts and made it possible to complete our project with all accurate information. We also wish to thank our parents for their personal support and attention who inspired me to go my own way.

We would also like to extend our sincere gratitude to our Head of the Department **Dr. Varsha Patil** and our Principal **Dr. K Lakshmisudha** for their continuous support and encouragement.

We also would like to thank our other faculty members for providing us with all the required resources and references for the project.

Project Team

Sahil Powar

Pranav Sukale

Swayam Sawant

Tanay More

List of Figures

- Figure 3.1 Flowchart of Proposed System
- Figure 5.1 SATHash retrieval interface demonstrating query input, binary hash generation, and top-k similar satellite image retrieval results. The system displays similarity scores, label matches, and bit-level differences in Hamming space.
- Figure 5.2 SATHash retrieval results showcasing high similarity matches for an agricultural satellite image query. The system retrieves top-k images with consistent semantic labels and minimal Hamming distance differences.
- Figure 5.3 Visualization of SATHash retrieval results generated during model evaluation in Jupyter Notebook. Green borders indicate correct label matches, while red borders represent mismatches based on semantic labels.
- Figure 5.4 Evaluation of SATHash binary representations showing Hamming distance distribution and per-bit activation rates. The distribution peaks near the ideal midpoint, and activation rates are balanced across bits, indicating effective hash learning.

List of Abbreviations

SATHash	Satellite Image Hashing System
CNN	Convolutional Neural Network
CBAM	Convolutional Block Attention Module
FAISS	Facebook AI Similarity Search
NT-Xent	Normalized Temperature-scaled Cross Entropy Loss
SimCLR	Simple Framework for Contrastive Learning of Visual Representations
GELU	Gaussian Error Linear Unit
RGB	Red Green Blue
GeoTIFF	Geospatial Tagged Image File Format
CBIR	Content-Based Image Retrieval
SIFT	Scale-Invariant Feature Transform
SURF	Speeded-Up Robust Features
ORB	Oriented FAST and Rotated BRIEF
mAP	Mean Average Precision
NDCG@10	Normalized Discounted Cumulative Gain at Rank 10
Precision@K	Precision at K
PR-AUC	Precision-Recall Area Under Curve
AdamW	Adaptive Moment Estimation with Weight Decay
BigEarthNet-S2	BigEarthNet Sentinel-2 Dataset

Chapter 1

Introduction

1.1 Introduction

SATHash is an intelligent and scalable satellite image retrieval system designed to efficiently identify and retrieve visually similar image patches from large-scale multispectral geospatial datasets. In recent years, the rapid growth of satellite imagery has created a significant demand for efficient storage, processing, and retrieval mechanisms. Traditional image comparison techniques are computationally expensive and unsuitable for large-scale datasets, while most existing deep learning approaches rely on labeled data, making them difficult to scale to the vast archives of unlabeled satellite imagery available today. SATHash addresses these challenges by combining self-supervised deep learning with binary hashing-based similarity search, enabling fast and accurate retrieval of satellite images without the need for exhaustive comparisons or manual annotations.

Developed using Python, SATHash is built around a custom deep neural architecture — Spectral-HashNetV6 — trained entirely from scratch using self-supervised contrastive learning, specifically the NT-Xent loss from the SimCLR framework. Unlike systems that rely on pre-trained models designed for standard RGB imagery, SATHash natively processes Sentinel-2 satellite images with 10 spectral bands, capturing rich land-cover information such as vegetation health, water content, and soil composition. The system leverages PyTorch for model development and Streamlit for building an interactive and user-friendly web interface. Specialized libraries such as Rasterio support geospatial image formats, while NumPy and JSON are used for efficient storage and management of feature embeddings and metadata.

The core functionality of SATHash lies in its ability to convert multispectral satellite images into

compact 64-bit binary hash codes through a pipeline consisting of a deep residual encoder with Convolutional Block Attention Modules (CBAM), a projection head, and a hash head with Tanh-Sign quantization. Each encoder stage uses ResidualBlockProj units with skip connections and CBAM attention, applying both channel and spatial attention to focus on the most discriminative spectral and spatial features. A key design decision was the adoption of Group Normalization over Batch Normalization, which preserves per-image statistical uniqueness and prevents hash collapse. The model is optimized using a composite loss function combining NT-Xent contrastive loss, quantization loss, and independence loss, with training stability ensured through AdamW, CosineAnnealingLR scheduling, gradient clipping, and warmup-based regularization ramp-up. To accelerate retrieval, the system integrates FAISS with Hamming distance, performing nearest-neighbor searches in milliseconds even at scale, supported by a precomputation strategy that stores binary codes in advance for real-time query responses.

SATHash also incorporates a rigorous evaluation framework using a proper train/test/query/database split, assessing retrieval performance through mAP, NDCG@10, Precision-Recall AUC, and Precision@K metrics. Users can upload a query image through the Streamlit interface, and the system retrieves the top-K most similar images based on Hamming distance. Results are displayed across three dedicated tabs for retrieval, analytics, and index browsing, with hash visualizations, Hamming similarity bars, and label match indicators. The system is designed with modularity and scalability in mind, allowing future enhancements such as Vision Transformer integration, real-time data ingestion, and geo-spatial indexing. In essence, SATHash transforms traditional satellite image retrieval into a fast, efficient, and intelligent process by combining self-supervised deep hashing, attention-guided spectral feature extraction, and high-performance binary search, making it a practical solution for urban planning, environmental monitoring, agriculture analysis, and disaster management.

1.2 Motivation

With the rapid advancement of remote sensing technologies and the increasing availability of high-resolution multispectral satellite imagery, there is a growing need for efficient systems to store, process, and retrieve geospatial data. Traditional methods of analyzing satellite images often involve manual inspection or computationally expensive pixel-wise comparisons, which are not scalable for large datasets. Furthermore, most existing deep learning-based retrieval

systems depend on labeled training data, which is costly and impractical to obtain at the scale of modern satellite archives. As the volume of satellite data continues to expand across multiple spectral bands, these limitations make it increasingly difficult to quickly identify relevant patterns or similar regions, especially in time-sensitive applications such as disaster monitoring and environmental analysis.

SATHash was developed to address these challenges by providing a fast and efficient satellite image retrieval system based on self-supervised deep learning and binary hashing techniques. The motivation behind training from scratch on 10-band Sentinel-2 imagery stems from the fact that standard pre-trained models are designed for RGB data and fail to capture the rich spectral signatures present in multispectral satellite data. Instead of directly comparing raw images, SATHash converts them into compact 64-bit binary hash codes that capture essential spatial and spectral characteristics, enabling ultra-fast Hamming distance-based similarity search without requiring any manual annotations.

Leveraging Python’s powerful ecosystem, including PyTorch for model development, FAISS for high-speed binary similarity search, and self-supervised contrastive learning for label-free training, SATHash transforms complex geospatial data into an accessible and scalable solution. The integration of Streamlit further enhances usability by providing an intuitive interface for uploading images and visualizing results. Ultimately, SATHash aims to overcome the inefficiencies of traditional satellite image analysis by combining spectral richness, self-supervised learning, and efficient binary retrieval — enabling faster decision-making and supporting applications such as urban planning, agriculture monitoring, environmental tracking, and disaster response through intelligent, data-driven insights.

1.3 Problem Statements and Objectives

The SATHash project aims to address the challenge of efficient retrieval of visually similar satellite images from large-scale geospatial datasets by providing a fast, scalable, and intelligent image similarity search system. With the exponential growth of satellite imagery, traditional approaches based on manual inspection or direct pixel comparison become computationally expensive and impractical. Additionally, most existing systems require labeled data for training, limiting their scalability to real-world unlabeled satellite archives. SATHash introduces a structured solution that combines self-supervised deep learning with binary hashing and high-

performance similarity search to enable accurate and rapid retrieval of relevant satellite image patches without any dependence on annotations.

The core objective of the system is to transform high-dimensional multispectral satellite images into compact 64-bit binary hash codes that can be efficiently compared using Hamming distance. Unlike systems that rely on pre-trained RGB models, SATHash trains SpectralHashNetV6 entirely from scratch on 10-band Sentinel-2 imagery, extracting discriminative embeddings that preserve essential spatial and spectral characteristics through a deep residual encoder augmented with Convolutional Block Attention Modules (CBAM). A key objective was also to solve hash collapse — ensuring that the generated binary codes are diverse, balanced, and semantically meaningful — achieved through Group Normalization and a composite loss function combining NT-Xent contrastive loss, quantization loss, and independence loss.

SATHash focuses on optimizing retrieval performance while maintaining accuracy and scalability. It incorporates techniques such as precomputation of binary hash codes, image patch generation for improved granularity, and the use of FAISS with Hamming distance for high-speed binary similarity search. A further objective is to establish a rigorous and fair evaluation framework by partitioning the dataset into dedicated training, test, query, and database splits, enabling unbiased measurement of retrieval performance through standard information retrieval metrics including Mean Average Precision (mAP), Normalized Discounted Cumulative Gain at rank 10 (NDCG@10), Precision-Recall AUC, and Precision at K for K values of 1, 5, 10, and 20. The system also provides an interactive Streamlit interface that allows users to upload query images and visualize the top-K matching results, making it accessible even to users without advanced technical expertise.

The project is designed with extensibility and future enhancements in mind, including the integration of advanced deep learning architectures such as Vision Transformers, real-time data processing, and geo-spatial indexing for location-aware retrieval. Additional objectives include enabling multi-modal search capabilities, supporting temporal analysis for change detection, and improving retrieval accuracy through model refinement. Overall, SATHash aims to provide a robust and forward-looking solution for efficient satellite image retrieval, supporting applications such as environmental monitoring, urban development, agriculture analysis, and disaster management.

1.4 Organization of the report

This report is organized to provide a structured overview of the design, development, and implementation of the SATHash satellite image retrieval system. The sections are as follows:

1. **Abstract:** Summarizes the purpose, key features, and technical approach of SATHash, highlighting its role as a self-supervised deep learning-based satellite image retrieval system that integrates binary hashing, contrastive learning, attention-enhanced feature extraction using SpectralHashNetV6, and high-speed similarity search using FAISS.
2. **Introduction:** Introduces the project, explaining the motivation behind SATHash and the need for efficient retrieval mechanisms for large-scale multispectral satellite imagery. It describes how the system leverages self-supervised contrastive learning, residual learning with CBAM attention mechanisms, 64-bit binary hashing, and high-performance similarity search to enable fast and accurate image retrieval without relying on labeled data.
3. **Problem Statement and Objectives:** Defines the challenges associated with handling large volumes of multispectral satellite data, including computational inefficiency, scalability issues, and dependence on labeled data. It outlines the objectives of SATHash, such as training SpectralHashNetV6 from scratch, preventing hash collapse using Group Normalization, incorporating attention mechanisms for improved feature extraction, and implementing a proper train/test/query/database split for fair evaluation.
4. **Design Details:** Covers the architectural design of the system, including data flow, system components, and overall pipeline. This section explains how multispectral images are processed using a residual attention-based encoder, converted into 64-bit binary hash codes, indexed, and retrieved, along with the design of the user interface built using Streamlit.
5. **Methodology:** Describes the development process, including data preparation on the BigEarthNet-S2 dataset, self-supervised training of SpectralHashNetV6 using NT-Xent contrastive loss, quantization loss, and independence loss, along with advanced optimization techniques such as AdamW, cosine annealing learning rate scheduling, warmup strategy, and gradient clipping. It also explains preprocessing steps and system workflow.
6. **Algorithm Implementation:** Explains the core algorithms used in SATHash, including residual convolutional blocks with skip connections, CBAM attention modules, GELU-based projection, Tanh-Sign binary quantization, and nearest-neighbor search using FAISS with Hamming distance. It details how binary hash codes are generated, indexed, and compared to retrieve

the most similar satellite images efficiently.

7. **Testing and Evaluation:** Describes the testing strategies used to ensure system reliability and performance, including validation of hash uniqueness, bit balance, mean Hamming distance, and evaluation using advanced metrics such as mAP, NDCG@10, Precision@K, and Precision–Recall AUC based on a proper query-database split.

8. **Conclusion and Future Scope:** Summarizes the achievements of SATHash in enabling efficient self-supervised satellite image retrieval using attention-enhanced deep hashing and discusses potential future enhancements, such as integration of Vision Transformers, real-time data processing, geo-spatial indexing, and multi-modal search capabilities.

9. **References:** Lists all tools, libraries, and resources used during the development of the project, including PyTorch, FAISS, Streamlit, NumPy, Rasterio, BigEarthNet-S2, SimCLR, CBAM, and other relevant frameworks and research works.

Chapter 2

Literature Survey

Before developing SATHash, a comprehensive review of existing satellite image retrieval and image similarity systems was conducted to understand current approaches, their strengths, and limitations. With the rapid growth of geospatial data, several systems and frameworks have been developed to manage and analyze satellite imagery. However, many of these systems face challenges related to scalability, efficiency, and intelligent retrieval. The following is an overview of some prominent solutions in this domain.

Google Earth Engine is a powerful cloud-based platform widely used for processing and analyzing large-scale geospatial data. It provides access to vast satellite datasets and supports complex spatial analysis using scripting. While it excels in large-scale computation and visualization, it is not specifically designed for similarity-based image retrieval. Users must manually define queries and analysis pipelines, making it less suitable for quick visual similarity search.

OpenCV-based Image Retrieval Systems are commonly used for basic image similarity tasks using feature descriptors such as SIFT, SURF, or ORB. These methods are effective for small datasets and simple applications. However, they struggle with high-dimensional multispectral satellite imagery and lack the ability to capture complex semantic features, resulting in lower accuracy and poor scalability for large geospatial datasets.

Content-Based Image Retrieval (CBIR) Systems utilize visual features such as color, texture, and shape to retrieve similar images. While CBIR techniques improve over traditional methods, they often rely on handcrafted features and do not fully leverage deep learning. As a result, their performance is limited when dealing with complex satellite imagery that requires deeper semantic understanding.

SimCLR (Chen et al., 2020) introduced a simple yet powerful framework for self-supervised contrastive learning using the NT-Xent loss, enabling models to learn meaningful visual represen-

tations without labeled data. However, SimCLR produces continuous floating-point embeddings and does not incorporate binary hashing, making it unsuitable for fast large-scale retrieval. SATHash builds directly on this foundation by adding a binary hash head and quantization mechanism on top of the contrastive learning framework.

HashNet (Cao et al., 2017) proposed learning binary hash codes directly through deep networks using a continuation method to handle the gradient issues of the sign function. However, HashNet relies on supervised learning and requires labeled image pairs or classes, limiting its scalability to unlabeled satellite archives. SATHash achieves comparable hashing quality in a fully self-supervised manner.

Deep Hashing Network (DHN, Zhu et al., 2016) jointly learns feature extraction and hash codes using a supervised framework with pairwise labels. While effective for standard image datasets, it is not designed for multispectral data and requires expensive annotation. SATHash eliminates this requirement through contrastive self-supervised training on 10-band Sentinel-2 imagery.

CBAM: Convolutional Block Attention Module (Woo et al., 2018) introduced a lightweight yet effective attention mechanism that sequentially applies channel attention and spatial attention to convolutional feature maps. Channel attention uses both average-pooled and max-pooled representations to identify which spectral channels are most informative, while spatial attention highlights the most relevant spatial regions. CBAM integrates seamlessly into standard CNN architectures with minimal overhead. SATHash incorporates CBAM into every stage of the SpectralHashNetV6 encoder through the ResidualBlockProj module, enabling the model to selectively focus on discriminative spectral bands and spatial structures within each satellite patch, directly improving the quality and separability of the learned binary hash codes.

Deep Contrastive Self-Supervised Hashing for Remote Sensing represents the closest existing work to SATHash, combining contrastive learning with hashing for satellite image retrieval. However, it operates on standard RGB remote sensing images and does not address hash collapse or bit decorrelation explicitly. SATHash improves upon this by incorporating Group Normalization, independence loss, residual connections, and CBAM attention within a strided convolutional encoder specifically designed for multispectral data.

Deep Learning-based Retrieval Systems in general use convolutional neural networks to extract feature embeddings from images, enabling more accurate similarity comparisons. These systems significantly outperform traditional approaches in terms of accuracy. However, many

implementations lack efficient indexing and search mechanisms, operate only on RGB data, and depend on labeled datasets, making them computationally expensive and difficult to scale.

From this review, several gaps were identified. Most existing systems either focus on large-scale processing without efficient similarity retrieval, require labeled data for training, or operate exclusively on RGB imagery without leveraging the rich spectral information available in multispectral satellite data. Few solutions effectively combine self-supervised learning with binary hashing, attention-guided feature extraction, and fast indexing to enable real-time retrieval at scale. SATHash addresses these limitations by integrating self-supervised contrastive learning, 64-bit binary hashing, CBAM attention, Group Normalization, and FAISS-based Hamming distance search into a unified pipeline trained on 10-band Sentinel-2 imagery, offering a practical, scalable, and annotation-free solution for satellite image retrieval.

2.1 Research Gaps

Although significant advancements have been made in satellite image processing and retrieval systems, many existing solutions still exhibit notable limitations. Most traditional systems rely on manual analysis or basic feature-based techniques such as SIFT, SURF, or ORB, which are computationally expensive and inefficient when dealing with large-scale multispectral geospatial datasets. Even with the adoption of deep learning, many platforms such as HashNet and DHN focus primarily on supervised feature extraction, requiring labeled data that is costly and impractical to obtain at the scale of modern satellite archives. This dependence on annotations severely limits their scalability to real-world unlabeled satellite imagery.

Additionally, most existing deep hashing approaches are designed for standard RGB imagery and fail to account for the rich spectral information present in multispectral satellite data such as Sentinel-2's 10 spectral bands. Several systems also suffer from hash collapse — a critical failure mode where all images are mapped to identical or near-identical binary codes — due to poor normalization choices such as Batch Normalization and inadequate loss design. This renders their binary representations useless for meaningful retrieval. Furthermore, many platforms lack efficient indexing mechanisms for high-dimensional binary vectors and do not support intuitive similarity-based retrieval, requiring users to define complex queries or workflows, restricting their accessibility to domain experts.

SATHash addresses these gaps by training SpectralHashNetV3 entirely from scratch using

self-supervised contrastive learning on 10-band Sentinel-2 imagery, eliminating the need for labeled data entirely. The adoption of Group Normalization over Batch Normalization, combined with a composite loss function comprising NT-Xent contrastive loss, quantization loss, and independence loss, directly solves hash collapse and ensures diverse, balanced 64-bit binary representations. FAISS-powered Hamming distance search enables millisecond-level retrieval at scale, while the interactive Streamlit interface simplifies user interaction. This combination of self-supervision, spectral richness, efficient binary indexing, and usability allows SATHash to overcome the limitations of existing systems and provide a practical solution for satellite image similarity retrieval across various applications.

2.2 Mini Project Contribution

The SATHash project contributes to the domain of satellite image analysis by introducing an efficient and scalable solution for similarity-based image retrieval. With the rapid increase in the availability of high-resolution multispectral satellite imagery, traditional retrieval methods based on manual inspection or direct image comparison have become impractical due to their high computational cost, dependence on labeled data, and lack of scalability. SATHash addresses these challenges by introducing a fully self-supervised deep hashing pipeline trained on 10-band Sentinel-2 imagery, eliminating the need for manual annotations while delivering fast and accurate retrieval.

One of the primary contributions of this project is the design and training of SpectralHashNetV6, a significantly upgraded custom deep neural architecture built from scratch to handle multispectral satellite data. Unlike existing approaches that rely on pre-trained RGB models or plain convolutional encoders, SpectralHashNetV6 introduces a deep residual encoder composed of ResidualBlockProj modules, each combining strided convolutions with skip connections and a Convolutional Block Attention Module (CBAM). CBAM sequentially applies channel attention, using both average-pooled and max-pooled features to identify the most informative spectral bands, followed by spatial attention to highlight the most discriminative spatial regions within each patch. This attention-guided encoding enables the model to produce far more separable and semantically meaningful binary representations compared to plain convolutional architectures. The use of Group Normalization over Batch Normalization remains a critical design decision, directly preventing hash collapse and ensuring each image produces a unique binary code.

Another key contribution is the composite loss function that drives the training process. By combining NT-Xent contrastive loss for semantic similarity learning, quantization loss for binary-friendly activations, and independence loss for bit diversity and decorrelation, the system achieves highly balanced and diverse 64-bit binary codes. Training stability is further enhanced through AdamW optimization with weight decay, CosineAnnealingLR scheduling, gradient clipping at max norm 1.0, and a warmup strategy that gradually ramps up regularization losses over the initial epochs, preventing early instability during contrastive learning.

A further contribution is the introduction of a rigorous evaluation framework based on a proper train/test/query/database split, enabling fair and unbiased assessment of retrieval performance. The system is evaluated using a comprehensive suite of information retrieval metrics including Mean Average Precision (mAP), Normalized Discounted Cumulative Gain at rank 10 (NDCG@10), Precision-Recall AUC, and Precision at K for K values of 1, 5, 10, and 20, all computed using ground-truth label overlap rather than index-based matching, ensuring meaningful and reproducible evaluation.

The integration of FAISS with Hamming distance enables efficient nearest-neighbor search in binary vector spaces, ensuring millisecond-level retrieval even as the dataset scales to millions of images. A precomputation strategy is also employed, where binary hash codes of the dataset are generated and stored in advance, significantly improving query response time during inference. The use of image patch generation further enhances retrieval granularity, allowing the system to identify similarities at a localized level rather than relying on entire images.

The project also emphasizes usability through a completely redesigned interactive Streamlit interface featuring three dedicated tabs for retrieval, analytics, and index browsing. The frontend supports both V3 and V6 model versions via a version selector, displays 64-bit hash codes as coloured grids with differing bits highlighted in red, shows Hamming similarity bars and label match indicators under each result, and reports search time in milliseconds. Overall, SATHash demonstrates the effective integration of attention-guided self-supervised deep hashing, residual spectral encoding, rigorous retrieval evaluation, and a modern web-based interface — providing a strong foundation for further advancements such as geo-spatial indexing, real-time data processing, and multi-modal search in intelligent geospatial analysis systems.

Chapter 3

Proposed System

3.1 System Architecture

The proposed SATHash system is a self-supervised deep learning-based satellite image retrieval platform designed to efficiently identify and retrieve visually similar multispectral satellite images from large-scale datasets. It addresses the limitations of traditional image retrieval methods, such as high computational cost, dependence on labeled data, lack of scalability, and inefficient similarity comparison. Built using Python, the system integrates PyTorch for model development, FAISS for high-speed binary similarity search, and Streamlit for an interactive frontend interface. By combining self-supervised contrastive learning, attention-enhanced deep feature extraction, 64-bit binary hashing, and optimized indexing, SATHash provides a fast, scalable, and annotation-free solution for satellite image similarity search.

Core Functionalities

SATHash enables efficient retrieval of similar satellite images by converting 10-band Sentinel-2 input images into compact 64-bit binary hash codes using SpectralHashNetV6, an enhanced deep neural architecture trained entirely from scratch. The model employs a residual convolutional encoder composed of *ResidualBlockProj* modules with skip connections, enabling deeper feature learning while preserving gradient flow. Each encoder stage is augmented with a Convolutional Block Attention Module (CBAM), which sequentially applies channel attention and spatial attention to refine feature representations by focusing on the most informative spectral and spatial regions. Group Normalization is used throughout the encoder to maintain per-image statistical consistency and prevent hash collapse. The encoder is followed by a projection head and a hash head, where GELU activation replaces ReLU for smoother optimization, and Dropout is applied

for regularization. The final output is a set of balanced and discriminative 64-bit binary hash codes generated via a differentiable Tanh-Sign quantization mechanism.

These hash codes are stored in a precomputed database to eliminate repeated computations during inference. The system utilizes FAISS with Hamming distance to perform fast nearest-neighbor search in binary vector space, significantly reducing retrieval time compared to traditional methods. Additionally, image patch generation is employed to divide large satellite images into smaller segments, improving retrieval precision and enabling localized similarity matching.

System Architecture

SATHash follows a modular architecture consisting of three main components: binary hash generation, indexing and search, and user interface. The hash generation module uses SpectralHashNetV6, trained with a composite loss function comprising NT-Xent contrastive loss, quantization loss, and independence loss, to convert multispectral input images into 64-bit binary codes. The training process is further stabilized using modern optimization strategies including AdamW optimizer, cosine annealing learning rate scheduling, warmup phases, and gradient clipping to ensure stable convergence and high-quality hash representations.

These binary codes are stored in a NumPy-based database along with metadata maintained in JSON format. The indexing and search module leverages FAISS with Hamming distance to efficiently index the binary codes and perform similarity search operations. The frontend is implemented using Streamlit, providing a responsive interface with advanced features such as model version selection, hash visualization, similarity metrics display, and real-time retrieval analytics. Data flows from the user input through the hash generation module to the FAISS index, and the retrieved results are mapped back to original images for visualization, ensuring a seamless and efficient pipeline.

Scalability and Future Roadmap

The proposed system is designed to overcome scalability issues by using compact binary representations and efficient Hamming distance indexing, allowing it to handle large satellite image datasets with minimal performance degradation. Since binary hash codes require significantly less memory than floating-point embeddings and XOR-based Hamming distance computation is extremely fast, the system can scale to millions of images without architectural changes. The improved architecture with attention mechanisms and residual learning further enhances representation quality without significantly increasing computational overhead.

Future enhancements may include integration of advanced architectures such as Vision Trans-

formers for improved spectral feature extraction, real-time data processing for dynamic datasets, and geo-spatial indexing for location-aware retrieval. Additional features such as multi-modal search (text-to-image), temporal analysis for change detection, and cloud-based deployment can further enhance system capabilities. The modular design ensures that new functionalities can be integrated without affecting existing components, making SATHash a robust and extensible solution for modern satellite image retrieval challenges.

3.2 Architecture of Proposed System

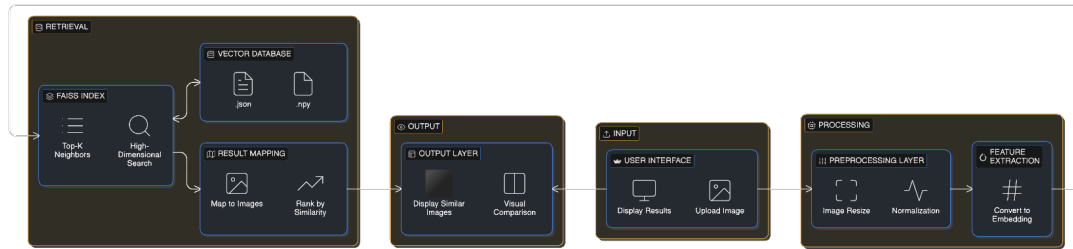


Figure 3.1: Flowchart of Proposed System

The architecture of SATHash follows a modular pipeline integrating self-supervised deep learning, binary hashing, efficient similarity search, and interactive visualization for multispectral satellite image retrieval. The top layer consists of a Streamlit-based user interface that allows users to upload query images, retrieve top-K matches, and visualize results in a structured and user-friendly manner.

Input images undergo preprocessing, including spectral normalization and augmentation of 10-band Sentinel-2 data to ensure consistency and improved representation learning. The processed inputs are passed to SpectralHashNetV6, a custom model with a residual encoder composed of *ResidualBlockProj* modules and CBAM attention (channel and spatial), along with Group Normalization to prevent hash collapse and preserve feature diversity.

Features are mapped to a 128-dimensional latent space via a projection head using GELU activation and Dropout, and then converted into compact 64-bit binary hash codes using a Tanh-Sign quantization mechanism. These codes are precomputed and stored in a NumPy database with metadata in JSON format, eliminating redundant computations during inference.

The FAISS-based search module performs fast similarity matching using Hamming distance

computed via XOR operations. The system retrieves the top-K nearest neighbors in milliseconds, ensuring scalability for large datasets.

Retrieved hash codes are mapped back to original images using metadata, ranked based on similarity, and displayed through the interface for easy comparison.

In summary, SATHash efficiently processes multispectral inputs into binary representations, performs fast Hamming distance search using FAISS, and delivers accurate retrieval results through an intuitive interface.

3.3 Details of Hardware and Software

The development and deployment of the SATHash system require a combination of hardware and software components to ensure efficient processing of multispectral satellite images, self-supervised model training, and fast binary similarity retrieval. The system is designed to operate on standard computing resources while also supporting scalability for larger datasets and enhanced performance.

Hardware Requirements: The SATHash system can be implemented on a standard personal computer with moderate specifications. A system with at least an Intel i5 processor (or equivalent), 16 GB RAM, and sufficient storage (minimum 50 GB) is recommended to handle the BigEarthNet-S2 dataset, precomputed binary hash codes, and model checkpoints efficiently. For training SpectralHashNetV6 from scratch using self-supervised contrastive learning, a dedicated GPU with CUDA support (such as an NVIDIA GTX 1650 or higher) is strongly recommended, as training on 10-band multispectral data with NT-Xent loss is computationally intensive. However, for inference and retrieval only, the system can also run on CPU-based environments with slightly higher processing time. Additional storage may be required depending on the size of the satellite image dataset and the number of precomputed hash codes stored.

Software Requirements: The SATHash system is developed using Python as the primary programming language due to its extensive support for machine learning and data processing libraries. PyTorch is used for designing, training, and running SpectralHashNetV3, including the strided convolutional encoder, Group Normalization layers, projection head, and Tanh-Sign hash head. FAISS (Facebook AI Similarity Search) is utilized for efficient binary indexing and Hamming distance-based similarity search. Streamlit is used to develop the interactive web-based user interface for image upload and result visualization.

Supporting libraries such as NumPy are used for numerical computations and storage of pre-computed binary hash codes, while JSON is used to manage metadata associated with images. Rasterio is employed for reading and preprocessing GeoTIFF satellite image formats from the BigEarthNet-S2 dataset. Additional libraries such as OpenCV or PIL may be used for image preprocessing tasks such as resizing and normalization. The self-supervised training pipeline also makes use of data augmentation utilities for generating multiple spectral views of each satellite patch during contrastive learning.

Development Environment: The system can be developed and executed using standard development tools such as Visual Studio Code or Jupyter Notebook. Training was performed on a GPU-enabled environment to accelerate the self-supervised contrastive learning process. The system is compatible with major operating systems including Windows, Linux, and macOS. Dependency management is handled through a requirements.txt file, ensuring easy installation and setup of all required libraries.

In summary, SATHash is designed to be flexible and efficient, with moderate hardware requirements for inference and higher configurations recommended for training from scratch on multispectral satellite data. The use of widely adopted software tools including PyTorch, FAISS, Rasterio, and Streamlit ensures ease of development, scalability, and efficient execution of the self-supervised satellite image retrieval system.

Chapter 4

Design and Methodology

4.1 Design details

The design of SATHash focuses on building an efficient, scalable, and user-friendly system for multispectral satellite image similarity retrieval. At the conceptual level, the system revolves around four core entities: the Query Image, SpectralHashNetV6, Binary Hash Database, and FAISS Index. The user provides a query satellite image, which is processed through the trained model to generate a 64-bit binary hash code. This code is then compared against precomputed binary hash codes stored in the database using Hamming distance, and the most similar images are retrieved. This design ensures a seamless flow from multispectral input to binary similarity-based output while maintaining high performance and accuracy.

For data design, SATHash utilizes NumPy arrays and JSON files as a lightweight and efficient storage solution. The 64-bit binary hash codes are stored in a .npy file for fast numerical operations, while metadata such as image paths, labels, and identifiers are stored in JSON format. This separation ensures efficient indexing, fast mapping, and scalability. Binary codes significantly reduce memory usage compared to floating-point embeddings, allowing the system to handle large datasets efficiently.

The Data Flow Diagram (DFD) represents the movement of data through the system. When a user uploads a query image, it undergoes spectral normalization and preprocessing to match the input requirements of SpectralHashNetV6. The processed 10-band image is passed through a residual encoder composed of *ResidualBlockProj* modules with skip connections, enhanced with CBAM attention (channel and spatial). The extracted features are passed through a projection head using GELU activation and Dropout, followed by a hash head with Tanh-Sign quantization

to produce a 64-bit binary hash code. This code is then compared with the FAISS-indexed database using Hamming distance, and the top-K similar results are retrieved and mapped back using metadata.

UML diagrams define the system structure with components such as SpectralHashNetV6, Preprocessor, FAISSIndexer, and ResultVisualizer. SpectralHashNetV6 handles feature extraction and hash generation using residual learning and attention mechanisms. The Preprocessor ensures consistent input formatting, the FAISSIndexer manages binary indexing and search, and the ResultVisualizer maps results and handles display. Sequence diagrams illustrate workflows including image upload, preprocessing, hash generation, binary search, and result visualization. The UI design emphasizes usability and clarity. The Streamlit interface allows users to upload images, retrieve results, and analyze outputs. It includes features such as model selection, hash visualization, similarity metrics, and search time display. Results are presented with the query image and top-K matches, making interpretation straightforward. The modular design supports future enhancements such as advanced analytics, geo-spatial visualization, and real-time data integration.

4.2 Methodology

To develop an efficient satellite image retrieval system, we began by identifying the key challenges associated with handling large-scale multispectral geospatial data, including high computational cost, slow retrieval speed, dependence on labeled data, and difficulty in comparing complex satellite images across multiple spectral bands. Traditional methods based on manual inspection or pixel-wise comparison were found to be inefficient and non-scalable, while existing deep learning approaches required expensive annotations. After analyzing existing approaches including SimCLR, HashNet, and DHN, it was concluded that a combination of self-supervised contrastive learning and binary hashing would provide the most practical and scalable solution. This led to the development of SATHash, a fully annotation-free system that integrates self-supervised deep hashing, binary indexing, and fast Hamming distance search for accurate and scalable satellite image retrieval.

The development process focused on four core components: data preparation, self-supervised model training, binary hash generation, and similarity search. The data preparation stage involved loading and preprocessing 10-band Sentinel-2 GeoTIFF patches from the BigEarthNet-

S2 dataset, applying spectral normalization and augmentation to generate multiple views of each image for contrastive learning. The upgraded SpectralHashNetV6 architecture was designed and trained entirely from scratch, incorporating a residual convolutional encoder composed of *ResidualBlockProj* modules with skip connections for improved gradient flow and deeper feature learning. Each encoder stage is enhanced with a Convolutional Block Attention Module (CBAM), which applies channel and spatial attention to refine feature representations by focusing on the most informative spectral bands and spatial regions.

The encoder is followed by a 128-dimensional projection head using GELU activation and Dropout for improved generalization, and a hash head with a differentiable Tanh-Sign quantization mechanism to produce compact 64-bit binary hash codes. The model is trained using a composite loss function comprising NT-Xent contrastive loss for semantic similarity learning, quantization loss to enforce binary-friendly activations, and independence loss to ensure bit diversity and decorrelation across the 64-bit code. To improve training stability and convergence, advanced optimization techniques are incorporated, including the AdamW optimizer with weight decay, CosineAnnealingLR scheduler for dynamic learning rate adjustment, gradient clipping to prevent instability, and a warmup strategy where regularization losses are gradually introduced using a scaling parameter.

Throughout the development process, emphasis was placed on solving hash collapse — a critical failure mode identified early in the project where all images were mapped to identical binary codes due to the use of Batch Normalization. Replacing Batch Normalization with Group Normalization, combined with the independence loss and improved architectural design using residual connections and attention mechanisms, proved to be the most impactful decision in maintaining stable and diverse hash representations. Continuous monitoring of hash quality metrics, including uniqueness and bit balance, was performed during training to ensure robustness.

The generated binary hash codes are stored in a NumPy-based database along with JSON metadata and indexed using FAISS with Hamming distance for fast nearest-neighbor retrieval. Precomputation of binary codes was implemented to reduce real-time processing overhead and improve query response time during inference. A proper dataset split strategy was adopted, dividing the data into training, testing, and further into query and database sets, ensuring fair evaluation and preventing data leakage.

The system was designed with a modular architecture, allowing each component — preprocessing, hash generation, binary indexing, and search — to function independently while maintaining

seamless integration. An enhanced Streamlit interface was developed to ensure ease of use, featuring real-time retrieval, similarity visualization, hash analysis, and performance metrics. Continuous testing and refinement were carried out to improve retrieval accuracy, hash quality, and system efficiency, supported by advanced evaluation metrics such as mAP, NDCG@10, Precision@K, and Precision–Recall AUC, along with millisecond-level retrieval performance.

4.3 Algorithm Implementation

Image Preprocessing and Binary Hash Generation Algorithm:

SATHash begins by transforming raw multispectral satellite images into compact binary hash codes suitable for similarity comparison. When a user uploads a query image, the system applies spectral normalization and resizing to the 10-band Sentinel-2 input to ensure consistency across spectral channels. The processed image is passed through SpectralHashNetV6, an enhanced deep neural architecture designed for multispectral feature learning.

The encoder consists of stacked *ResidualBlockProj* modules with skip connections, enabling efficient gradient propagation and deeper representation learning. Each residual block is augmented with a Convolutional Block Attention Module (CBAM), which applies channel attention followed by spatial attention to emphasize informative spectral bands and spatial regions while suppressing irrelevant features. Group Normalization is used throughout the network to maintain per-image statistical consistency and prevent hash collapse.

The extracted features are then passed through a projection head that maps them into a normalized 128-dimensional latent space using GELU activation, followed by Dropout for regularization. Finally, a hash head applies a differentiable Tanh-Sign quantization mechanism to generate compact and balanced 64-bit binary hash codes that preserve semantic similarity across images.

Binary Code Storage and Indexing Algorithm:

The system employs a precomputation strategy where 64-bit binary hash codes of all satellite image patches in the dataset are generated in advance using SpectralHashNetV6 and stored in a NumPy-based database. Along with hash codes, metadata such as image paths, labels, and identifiers are stored in a JSON file for efficient mapping and evaluation.

These binary codes are indexed using FAISS with Hamming distance, constructing an optimized data structure for fast binary similarity search. The use of compact binary representations significantly reduces memory requirements and enables efficient indexing even for large-scale

datasets. This approach eliminates redundant computations during inference and ensures rapid retrieval performance.

Similarity Search Algorithm:

For a given query image, the generated 64-bit binary hash code is compared against the indexed database using FAISS with Hamming distance, computed via efficient XOR bit operations. FAISS retrieves the top-K nearest neighbor codes based on minimum Hamming distance, ensuring that the most semantically similar images are returned.

The retrieved results are ranked according to similarity scores derived from Hamming distance, enabling accurate and consistent similarity matching. This binary search approach ensures millisecond-level retrieval performance, making the system suitable for large-scale satellite image repositories.

Result Mapping and Visualization Algorithm:

Once the top-K similar binary codes are identified, the system maps them back to their corresponding satellite image patches using metadata stored in the JSON file. The results are sorted based on similarity ranking and prepared for visualization.

The Streamlit interface displays the query image alongside the retrieved results, along with additional analytics such as Hamming similarity scores, label match indicators, and hash visualizations that highlight differing bits between the query and retrieved images. This enhances interpretability and provides deeper insight into the retrieval process.

Training Optimization and Adaptive Processing:

SATHash incorporates advanced training strategies to ensure stable convergence and high-quality binary representations. The model is trained using the AdamW optimizer with weight decay for better generalization, combined with a CosineAnnealingLR scheduler to gradually adjust the learning rate during training. A warmup strategy is employed in the initial epochs, where regularization losses are gradually introduced using a scaling parameter, allowing the model to first learn stable representations before enforcing strict binary constraints.

Gradient clipping is applied during training to prevent exploding gradients and stabilize optimization, particularly in deeper residual architectures. The composite loss function includes NT-Xent contrastive loss for semantic similarity learning, quantization loss for enforcing binary-friendly activations, and independence loss for ensuring bit diversity and decorrelation.

During inference, precomputed binary hash codes and FAISS-based indexing significantly reduce computational overhead and improve response time. The system supports incremental updates,

allowing new images to be processed, hashed, and added to the database without requiring full re-indexing. This ensures scalability and adaptability for continuously growing satellite image datasets.

Chapter 5

Result and Discussions

5.1 Model Implementation

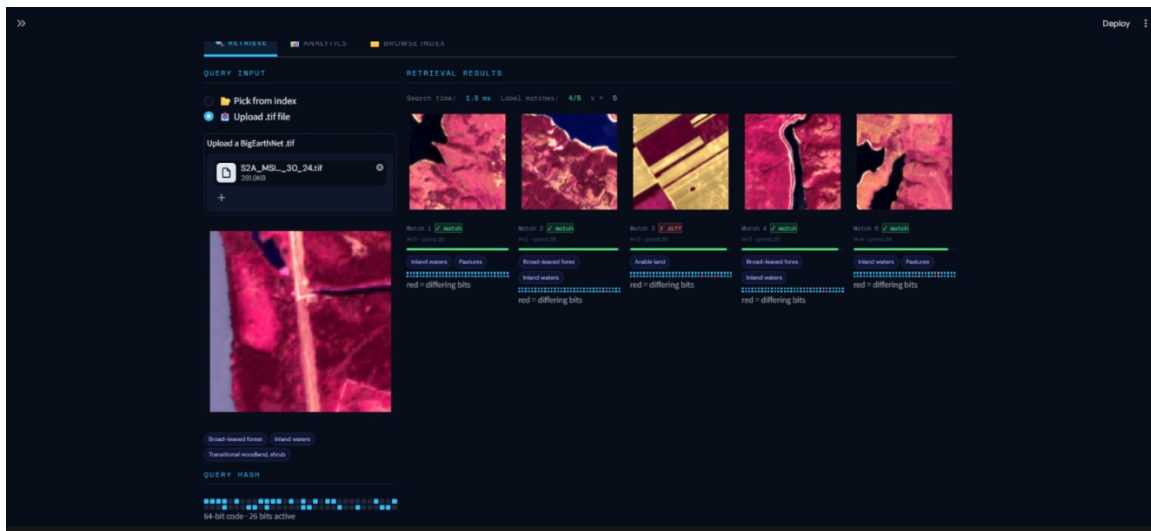


Figure 5.1: SATHash retrieval interface demonstrating query input, binary hash generation, and top-k similar satellite image retrieval results. The system displays similarity scores, label matches, and bit-level differences in Hamming space.

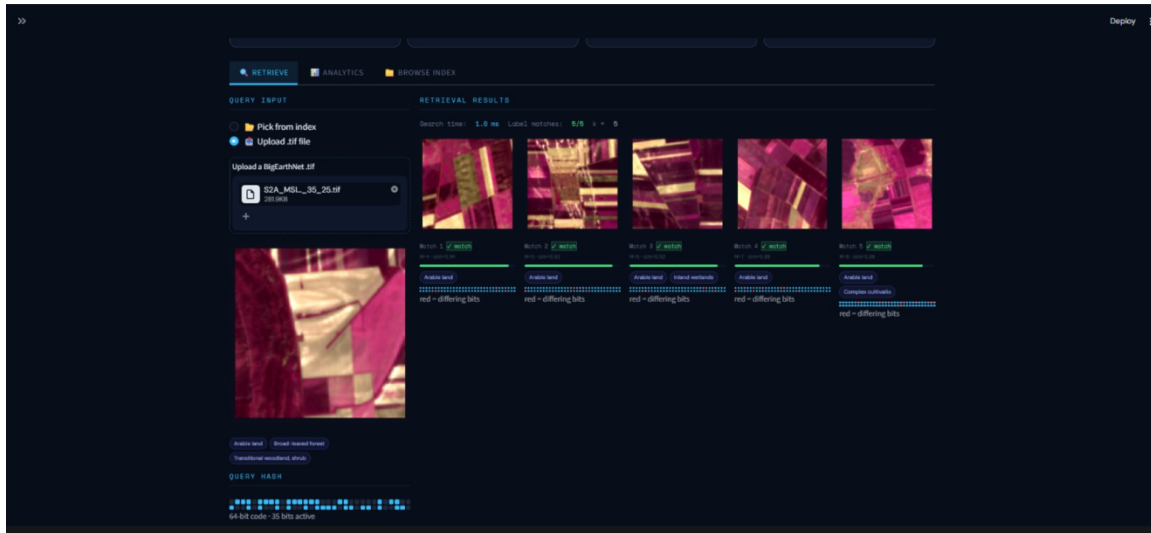


Figure 5.2: SATHash retrieval results showcasing high similarity matches for an agricultural satellite image query. The system retrieves top-k images with consistent semantic labels and minimal Hamming distance differences.

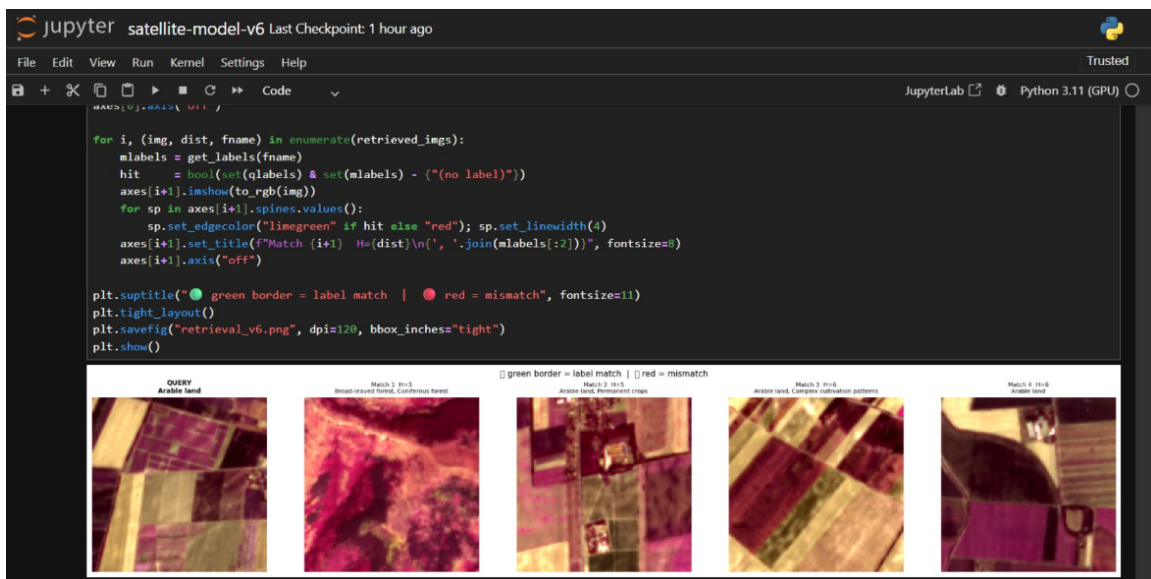


Figure 5.3: Visualization of SATHash retrieval results generated during model evaluation in Jupyter Notebook. Green borders indicate correct label matches, while red borders represent mismatches based on semantic labels.



Figure 5.4: Evaluation of SATHash binary representations showing Hamming distance distribution and per-bit activation rates. The distribution peaks near the ideal midpoint, and activation rates are balanced across bits, indicating effective hash learning.

5.2 Results and Discussions

The implementation of SATHash demonstrated significant improvements in both retrieval performance and evaluation rigor with the introduction of SpectralHashNetV6. The upgraded architecture, incorporating residual learning and CBAM-based attention mechanisms, enabled more effective feature extraction from 10-band Sentinel-2 imagery. The system was evaluated using a proper dataset split, where the data was divided into training, testing, and further into query and database sets, ensuring fair and unbiased performance assessment. This setup eliminated data leakage and provided a realistic evaluation of retrieval capability.

The quality of the generated 64-bit binary hash codes was validated through multiple internal metrics. The system achieved a high percentage of unique hash codes across the dataset, confirming that hash collapse was successfully avoided. All bits exhibited balanced activations, with the mean Hamming distance remaining close to the theoretical ideal of 32, indicating that the learned binary representations are diverse, decorrelated, and information-rich. The combination of Group Normalization, CBAM attention, residual connections, and the composite loss function played a critical role in achieving these stable and high-quality representations.

Retrieval performance was evaluated using standard information retrieval metrics, providing a comprehensive assessment beyond basic precision measures. The system achieved strong

results in Mean Average Precision (mAP), demonstrating its ability to retrieve relevant images consistently across multiple queries. Additionally, the Normalized Discounted Cumulative Gain at rank 10 (NDCG@10) indicated high ranking quality, ensuring that the most relevant results appear at the top of the retrieval list. Precision@K metrics, including Precision@1, Precision@5, Precision@10, and Precision@20, were evaluated in a label-based manner, confirming the model's effectiveness in capturing semantic similarity across multispectral inputs.

The Precision–Recall curve further illustrated the trade-off between precision and recall across different thresholds, and the corresponding Area Under Curve (PR-AUC) demonstrated the robustness of the retrieval system. Compared to earlier versions, the inclusion of these evaluation metrics provides a more reliable and research-aligned performance analysis, making the system suitable for benchmarking against existing methods.

The FAISS-based Hamming distance search enabled millisecond-level retrieval, ensuring scalability even for large datasets. The use of binary hash codes significantly reduced memory usage and computational overhead compared to floating-point embeddings. The Streamlit-based interface enhanced usability by providing real-time retrieval, similarity scores, search time display, and hash visualizations, allowing users to interpret results more effectively. The visualization of differing bits between query and retrieved images provided additional insight into the binary similarity mechanism.

The integration of image patch generation improved retrieval granularity by enabling localized similarity detection within large satellite images. The system consistently retrieved semantically similar regions such as urban areas, vegetation zones, and water bodies, demonstrating the effectiveness of multispectral feature learning enhanced by attention mechanisms.

Despite these advancements, certain limitations remain. Retrieval performance may vary for highly ambiguous or spectrally complex regions where class boundaries overlap. Additionally, while the current system uses a well-defined query-database split for evaluation, real-time dynamic updates to the database are still limited. Future work may focus on incorporating advanced architectures such as Vision Transformers, expanding dataset diversity, enabling real-time indexing, and integrating geo-spatial metadata to further enhance retrieval accuracy and contextual understanding.

Chapter 6

Conclusion

6.1 Conclusion

SATHash has proven to be an effective and intelligent satellite image retrieval system designed to address the challenges of handling large-scale multispectral geospatial data. By introducing a fully self-supervised deep hashing pipeline trained entirely from scratch on 10-band Sentinel-2 imagery, the system eliminates the dependence on labeled data that limits the scalability of most existing approaches. The upgraded SpectralHashNetV6 architecture incorporates residual learning and CBAM-based attention mechanisms, enabling more effective extraction of spatial and spectral features. The combination of *ResidualBlockProj* modules, channel and spatial attention, and improved projection and hash heads using GELU activation and Dropout allows the model to generate compact and highly discriminative 64-bit binary hash codes.

The composite loss function comprising NT-Xent contrastive loss, quantization loss, and independence loss ensures that the generated codes are semantically meaningful, diverse, and balanced. The system maintains a high percentage of unique hash codes, balanced bit activations, and a mean Hamming distance close to the theoretical ideal, confirming the effectiveness of the learned binary representations. In addition to internal validation metrics, the system was evaluated using a proper train, test, query, and database split, ensuring fair and reliable performance assessment. Retrieval effectiveness was further validated using advanced metrics such as Mean Average Precision (mAP), Normalized Discounted Cumulative Gain at rank 10 (NDCG@10), Precision@K, and Precision–Recall AUC, demonstrating strong semantic retrieval performance across multiple evaluation criteria.

The integration of FAISS with Hamming distance search enables millisecond-level retrieval at

scale, while the precomputation strategy eliminates redundant forward passes during inference. The use of binary hash codes significantly reduces memory and computational overhead, allowing the system to scale efficiently to large datasets. The enhanced frontend interface further improves usability by providing real-time retrieval, similarity visualization, and analytical insights into hash behavior and retrieval quality.

Overall, SATHash delivers a practical, scalable, and annotation-free solution for satellite image similarity retrieval. By combining self-supervised learning, attention-enhanced deep hashing, and efficient binary search, the system bridges the gap between traditional image comparison methods and modern geospatial analysis techniques. It provides a strong foundation for future enhancements such as Vision Transformer integration, geo-spatial indexing, real-time data processing, and multi-modal search capabilities, ultimately contributing to more intelligent and efficient handling of satellite imagery.

6.2 Future Scope

SATHash offers significant potential for further enhancement and expansion. One important area is real-time data integration, allowing the system to dynamically update the binary hash database as new Sentinel-2 imagery becomes available, ensuring retrieval results remain relevant for time-sensitive applications such as disaster monitoring and environmental tracking.

Another key direction is the integration of advanced architectures such as Vision Transformers (ViT) or spectral attention mechanisms, which can better capture long-range spatial dependencies and complex multispectral patterns. Extending the hash code length beyond 64 bits could also improve retrieval precision for larger and more diverse datasets. Incorporating geo-spatial metadata such as coordinates and temporal information would further enable location-aware and time-based search.

The system can also be extended to support multi-modal search capabilities, where users query images using textual descriptions in addition to image inputs. Cloud-based deployment and distributed FAISS indexing would improve scalability, allowing the system to handle millions of images and support multiple users simultaneously.

Finally, improvements in visualization such as spectral band viewers, map-based interfaces, and real-time comparison tools would enhance user experience. Integration with external GIS platforms and support for mobile interfaces would further increase accessibility, transforming

SATHash into a more powerful and intelligent platform for geospatial data analysis.

Bibliography

- [1] Meta AI. Faiss github repository, 2025. <https://github.com/facebookresearch/faiss>.
- [2] Various Authors. Concept-based content-based image retrieval for geospatial data. *Information Sciences*, 2025. CBIR with FAISS for satellite imagery.
- [3] Zhangjie Cao, Mingsheng Long, Jianmin Wang, and Philip S. Yu. Hashnet: Deep learning to hash by continuation. *arXiv*, 2017. <https://arxiv.org/abs/1702.00758>.
- [4] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. *arXiv*, 2020. ICML 2020. <https://arxiv.org/abs/2002.05709>.
- [5] Wei Chen, Yu Liu, and Weiping Wang. Deep learning for instance retrieval: A survey. *arXiv*, 2021. Survey on deep learning retrieval techniques.
- [6] GitHub Community. Content-based image retrieval using faiss, 2023. <https://github.com/KhaLee2307/image-retrieval>.
- [7] NumPy Developers. *NumPy Documentation*, 2025. <https://numpy.org/doc/>.
- [8] Rasterio Developers. *Rasterio Documentation*, 2025. <https://rasterio.readthedocs.io/>.
- [9] Albert Gordo, Jon Almazan, Jerome Revaud, and Diane Larlus. Deep convolutional learning for content based image retrieval. *Neurocomputing*, 2018. CNN-based feature extraction for image retrieval.
- [10] Jeff Johnson, Matthijs Douze, and Herve Jegou. Billion-scale similarity search with gpus. *IEEE Transactions*, 2017. FAISS large-scale vector search.

- [11] Jeff Johnson, Matthijs Douze, and Herve Jegou. *FAISS: Facebook AI Similarity Search Library*, 2017. <https://faiss.ai/>.
- [12] Xiaoqing Li et al. Recent developments in content-based image retrieval. *Neurocomputing*, 2020. Survey on CBIR techniques and deep learning approaches.
- [13] Subhadip Maji and Smarajit Bose. Cbir using features derived by deep learning. *arXiv*, 2020. Deep feature embeddings for CBIR.
- [14] Gencer Sumbul, Jian Kang, and Begum Demir. Deep learning for image search and retrieval in large remote sensing archives. *arXiv*, 2020. Deep learning-based CBIR for satellite imagery.
- [15] Ahmad Tarawneh and Ceyhun Celik. Investigation of deep features with dimensionality reduction in cbir. *arXiv*, 2018. Comparison of deep features vs traditional features.
- [16] PyTorch Team. *PyTorch Documentation*, 2025. <https://pytorch.org/docs/stable/index.html>.
- [17] Streamlit Team. *Streamlit Documentation*, 2025. <https://docs.streamlit.io/>.
- [18] Sarath Chandra Yenigalla, Karumuri Srinivasa Rao, and Phalguni Ngangbam. Implementation of content-based image retrieval using artificial neural networks. *Engineering Proceedings*, 2023. CBIR using CNN, SVM and KNN techniques.
- [19] Han Zhu, Mingsheng Long, Jianmin Wang, and Yue Cao. Deep hashing network for efficient similarity retrieval. 2016. <https://bitsnnfl2020.github.io/cnn/ir/deep-hashing-network-for-efficient-similarity-retrieval/>.